Dangers of Radiation

Exam code: 0625 & 0972

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Dangers of radioactivity

- Ionising nuclear radiation can **damage living things** such as human cells and tissues at high doses:
- This can include:
  - Cell death
  - Mutations
  - Cancer

- If the atoms that make up a DNA strand are **ionised,** then the DNA strand can be **damaged**
- If the DNA is damaged, then the cell may die, or the DNA may be **mutated** when it is replicated
- If a mutated cell replicates itself then a **tumour** may develop
  - This is an example of **cancer**, which is a significant danger of radiation exposure

Ionising DNA is a danger of radioactivity

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

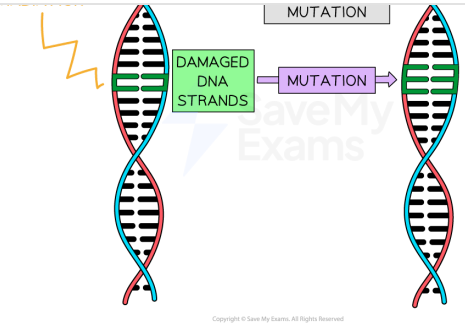


Diagram showing the damage caused to DNA by ionising radiation. Sometimes the cell is able to repair the DNA during replication successfully, but incorrect repairs can cause a mutation

- Acute radiation exposure can have other serious symptoms:
  - It can cause skin **burns**, similar to severe sunburn
  - Radiation can **reduce** the amount of **white blood cells** in the body, making a person more susceptible to infections by lowering their immune system
- Because of this, it is very important to handle radioactive sources carefully

Safe storage

- Radioactive materials are moved, used and stored in a safe way

Safe handling to minimise the dangers of radioactivity

- The risks associated with handling radioactive sources can be minimised by following a few simple procedures:
  - Store the sources in **lead-lined boxes** and keep them at a distance from people
  - **Minimise** the amount of **time** you handle sources and return them to their boxes as soon as you have finished using them
  - During use, keep yourself (and others) as **far** from the sources as possible.
  - When handling the sources do so at **arm's length**, using a pair of tongs

A sign indicating the dangers of radioactivity

Radioactivity warning sign

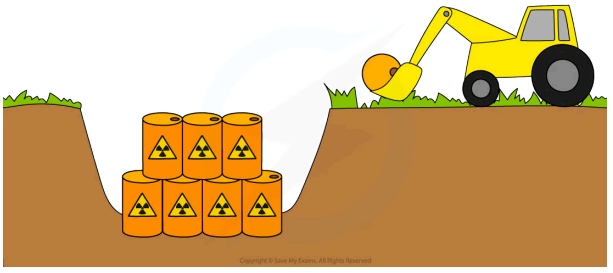
Safe transportation to minimise the dangers of radioactivity

- Radioactive materials such as used nuclear fuel are transported in special containers called casks
- These casks can withstand extreme conditions such as fire, cold and being submerged in water

Disposing of radioactive waste to minimise the dangers of radioactivity

- If an isotope has a long half-life then a sample of it will decay slowly
  - Although it may not emit a lot of radiation, it will **remain radioactive for a very long time**
- Sources with long half-life values present a risk of **contamination** for a much longer time
- Radioactive waste with a long half-life is **buried** underground to prevent it from being released into the environment

Correct disposal of radioactive waste reduces the dangers of radioactivity



Radioactive waste with long half-lives is buried deep underground

experiment. List four things that the student should do in order to minimise the risk to themselves when using the source.

Answer:

Any four from:

- Keep the source in a lead-lined container until the time it is needed
- Use tongs to move the source, rather than handling it directly
- The source should be kept at as far a distance from the student as possible during the experiment
- The time that the source is being used should be minimised
- After the experiment, the student should wash their hands
- The date and the time that the radiation has been used should be recorded

Safety precautions

Extended tier only

- Safety precautions for all ionising radiation include:
  - reducing **exposure time**
  - increasing the **distance** between the source and living tissue
  - using **shielding** to absorb radiation

Reducing exposure time reduces the dangers of radioactivity

- Limiting the amount of time spent near a radioactive source reduces the amount of radiation dose received
- The amount of radiation received by a person is called the dose and is measured in **sieverts** (Sv)
- One sievert is a very big dose of radiation
  - It would cause acute **radiation poisoning**

Increasing the distance reduces the dangers of radioactivity

- **Increasing** the **distance** the radiation is away greatly **reduces** the size of the **dose** received
  - Using tongs instead of your hand when handling a radioactive source can help with this

- Constructing nuclear power plants in **remote areas** increases their distance from people if there is a problem
- Burying nuclear waste far from places where people live also increases **the distance** to people

Shielding reduces the dangers of radioactivity

radiation at a certain location or time

- Barriers **absorb the energy** from radioactive sources
- Barriers are normally made of lead, water or concrete
  - People handling radioactive sources should wear a **lead apron** to reduce their exposure

- When X-rays are taken:
  - the area around where the image is needed is shielded
  - the radiologist stands behind a barrier or leaves the room

Reducing the dangers of radiation for a radiologist

A radiologist can stand behind a lead barrier to reduce the amount of radiation they are exposed to

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